TFG-121

Terafence
Reliable Transformations!

Technology Overview

Designed to protect IoT and IIoT equipment, Terafence **TFG-121** uses Air-Gap to segmentize and isolate industrial end devices from any harmful, or abusive, malware attacks and thus secure them from the network-based cyberattacks.

Terafence's proprietary hardware-architecture based on FPGA technology developed and manufactured by Terafene, creates a fully controlled data path between two network segments and while allowing normal protocol data to flow from one side to the other, the return path simply does not exist, hence Air-Gapped.

Terafence is acting as a Protocol Proxy, terminating TCP/IP sessions on both ends and only moving raw data between the two unidirectional gateway sides. RAW data is not stored within the unit thus eliminating the requirement to safeguard such data by encryption or other methods. As nothing is stored, no such tools are used (like data encryption, compression or alike). Terafence technology and network mechanisms do not use cryptology to secure data exchange but instead denies network access.

Terafence **TFG-121** not only protects IoT/IIoT end-devices from cyberattacks but also can protect other network assets by blocking any malicious attempts by an already infected external end-devices to infiltrate and infect devices with malicious code or cause other kind of damage.

Supported Protocols
TFG-121 supports multiple, simultaneous industrial and IT protocols, including:

- **SFTP/FTP/SMB –** Secure file or folder-tree transfers
- **SMTP –** Email relay
- **SYSLOG Forwarding –** Log event transmission
- **RTSP/ONVIF –** Live streaming for CCTV
- **HTTP/S –** File and data uploads to web servers or cloud platforms
- **Local SYSLOG Forwarding –** Internal log event management
- **Modbus –** Communication between multiple PLCs and HMIs
- **IEC-104 –** Protocol for power utilities
- **MQTT –** Data brokering and publishing
- And various other protocols as per customer requirements

Why choose TFG-121?

Key Features:

- Complete galvanic **network isolation** for enhanced security and protection.
- Proprietary **Terafence FPGA-based hardware** architecture for **hardware-enforced security.**
- Two dedicated CPUs for **protocol support, processing, and termination.**
- **Secure HTTPS-based web GUI** for configuration and management, accessible only from the secure side.
- **Hardened** operating system on the dedicated CPUs.
- Core security hardware operates independently **without** an operating system, MAC address, or IP address.

Technical Specifications:

- **Data Throughput:** Up to 1 Gbps with auto-negotiation
- **Power Supply:** External 12-48VDC @ 60 Watt (2 inputs for redundancy)
- **Network Interfaces:** 2 × Ethernet (RJ-45) ports
- **Form Factor:** DIN Rail Mount
- **Dimensions:** 95 x 160 x 200 (mm)
- **Operating Temperature:** -40 °F to 185 °F
- **Storage Temperature:** -40 °F to 185 °F
- **Design:** Fanless architecture with no moving mechanical parts for enhanced reliability
- **Compliance:** FCC, CE, and EMI/EMC

Solution Highlights:

- Delivers **physical network isolation** through Terafence's proprietary **FPGA-based** hardware architecture.
- Supports **protocol processing** and termination through two dedicated proxy engines.
- Provides secure and intuitive configuration through a restricted **HTTPS**-based management interface.
- Designed for high availability, long-term operation, and a **high MTBF.**
- Certified to **IEC 62443** and **MIL-STD-810F** standards.

All images are for illustrative purposes only. Product configurations, specifications, and features may be customized to meet customer-specific requirements and may vary accordingly.

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